

Visakh Padmanabhan

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LinkedIn | GitHub | Portfolio | Medium

Profile Summary

AI & Software Engineer with three years of experience in Germany building and **productionizing end-to-end AI and Machine Learning systems: LLM-based agentic application development, RAG optimization, evaluation pipelines, and production deployment.** **Best Outgoing Student** (M.S., **Grade 1.4**, Otto von Guericke University) with a **JURIX 2024 Best Student Paper** in legal AI and NLP. **EU Blue Card** holder (full work authorization), open to relocation.

Professional Experience

AI & Software Engineer

January 2026 – Present

MaindTec GmbH, Ingolstadt, Germany

- Core developer on **MaiQ**, an AI-powered project management tool, shipping **end-to-end ML features** from document processing to intelligent retrieval into **production**.
- Built MaiQ's **Skill Store**, letting end users (not only engineers) author, share, and run modular **agent capabilities** directly in chat, broadening who can extend the product.
- Cut **inference cost 37%** (with only a 2% drop in retrieval quality) by running **11+ retrieval experiments** under an **LLM-as-judge** evaluation framework and productionizing the best strategy.
- Improved **document relevance** for end users by rebuilding retrieval as **hybrid search** with **agentic, tool-calling retrieval** in **LangChain**.
- Built the **event-driven ingestion pipeline (Redis)** that powers MaiQ's **semantic search** over uploaded documents.

Junior Data Scientist

January 2025 – August 2025

Vay Technology, Berlin, Germany

- Cut **ETA prediction error 25%** through statistical evaluation of routing performance, sharpening logistics decisions.
- Built **dbt** monitoring models that surfaced operational issues early and gave stakeholders reliable mission performance reporting.

Data Scientist, Working Student

May 2022 – December 2024

Einbliq.io, Munich, Germany

- Developed **ML forecasting models** for streaming-device energy use to estimate **power emissions and carbon footprint**, shipping the supporting data service at **99% data coverage**.
- Built monitoring dashboards that gave the team visibility into streaming sessions and device performance.

Systems Engineer

October 2019 – August 2021

Infosys R&D Ltd, Thiruvananthapuram, India

- Built an **NLP pipeline (spaCy)** to detect and redact sensitive information from unstructured text via **named entity recognition**.

Technical Skills

- **Programming & Web:** Python, SQL, Next.js
- **LLM & Agentic:** LangChain, LangGraph, RAG optimization, LLM tool calling, agentic retrieval, LLM-as-judge evaluation
- **NLP & ML:** Sentence-Transformers (Sentence-BERT), Knowledge Graphs (Neo4j), Hugging Face, Scikit-learn, XGBoost, spaCy
- **Data & Backend:** PostgreSQL/pgvector, Redis, FastAPI, Flask, dbt, event-driven architecture
- **Cloud & DevOps:** AWS, Azure, Docker, CI/CD, GitHub Actions, Git
- **Languages:** Malayalam (Native), English (C1), German (A2)

Education

M.S. in Data and Knowledge Engineering <i>Otto von Guericke University, Magdeburg, Germany</i>	<i>October 2021 – March 2025</i> Grade: 1.4
B.Tech in Computer Science and Engineering <i>Mar Baselios College of Engineering and Technology, Thiruvananthapuram, India</i>	<i>July 2015 – August 2019</i> Grade: 1.9

Projects and Publications

- **Hybrid Legal Norm Retrieval** – Leveraging Knowledge Graphs and Sentence-BERT for retrieving Civil Code Articles (COLIEE 2024 Task 3). *Awarded Best Student Paper, JURIX 2024.* [\[Link\]](#)
 - Designed a hybrid retrieval pipeline combining **knowledge graph traversal** with **Sentence-BERT** embeddings to retrieve relevant legal norms from the Japanese Civil Code, achieving competitive results on the COLIEE 2024 benchmark.
- **LUMI: Legal Understanding and Matching through Interactive Highlighting** – Improved legal text explainability by highlighting semantically aligned spans. [\[Link\]](#)
- **Expert Agent Guided Learning** – Chatbot for guided learning using knowledge graphs built from OCR-extracted textbook content. [\[Link\]](#)
- **Slide Recommendation System** – OCR-based keyword extraction and ranking system to enhance student learning from slides. [\[Link\]](#)
- **MindMate** – Conversational AI Agent prototype using **LangGraph**, **LangChain**, and OpenAI API to deliver empathetic, context-aware dialogue with personalized coping strategies. [\[Personal Project Link\]](#)

Achievements

- **Best Outgoing Student Award (M.S.)** – Otto von Guericke University, Magdeburg
- **Best Student Paper Award, JURIX 2024** – Hybrid Legal Norm Retrieval: Leveraging **Knowledge Graphs** and **Sentence-BERT** for Retrieving Civil Code Articles (COLIEE 2024 Task 3)
- **All Round Performance Award** – Best Outgoing Student (2015–2019)